Uber Traffic Volume Prediction - EDA Summary

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Introduction: Addressing Urban Traffic Congestion

This presentation outlines the process of developing a predictive model to forecast hourly traffic volumes at various road junctions. Our goal is to leverage historical data to understand key influencing factors and propose targeted interventions to alleviate urban congestion, a pervasive issue stemming from:

- Road capacity limitations and infrastructure bottlenecks.
- Traffic incidents, such as accidents or unexpected road closures.
- Ongoing road work and maintenance activities.
- High vehicle density, particularly exacerbated by the increasing rate of vehicle ownership in urban households.

Agenda

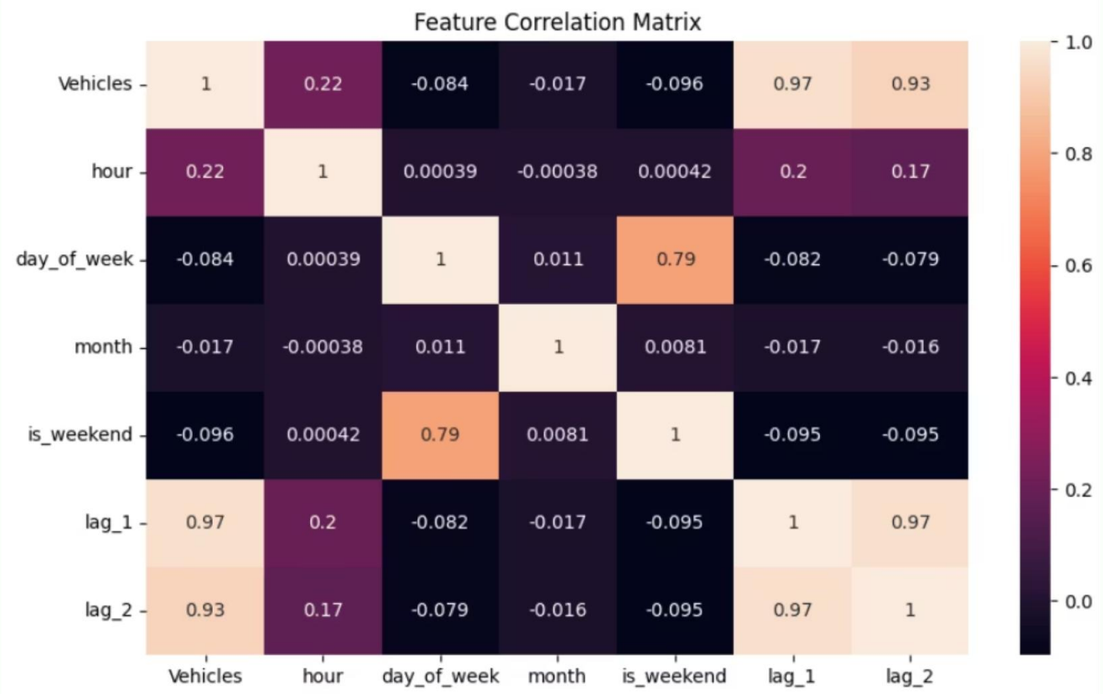
1. Problem Statement & Introduction
2. Key Predictors of Traffic Volume
3. Correlation Analysis
4. Random Forest Feature Importance
5. EDA-Based Modeling Recommendations
6. Summary

Strong Predictors: Lag Features

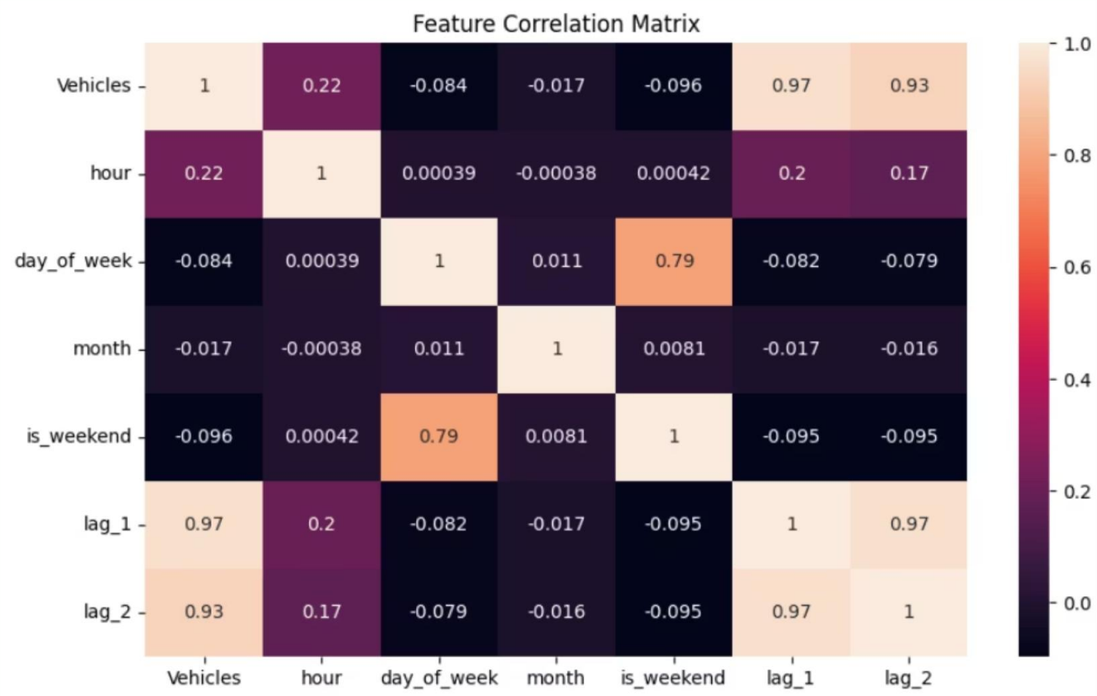
Our exploratory data analysis reveals that time-lagged traffic volumes are exceptionally strong predictors of current traffic conditions.

- **lag_1:** The traffic volume from the immediate preceding hour shows a correlation coefficient of **0.97** with the current volume.
- **lag_2:** Traffic volume from two hours prior also exhibits a very high correlation of **0.93**.

These strong correlations underscore the importance of incorporating these features to achieve high model accuracy for hourly traffic forecasts.



Time-Based Features: Weekend Impact



The analysis of time-based features highlights a significant distinction in traffic patterns between weekdays and weekends:

- **Day of Week vs. Is Weekend:** These features exhibit a high positive correlation of **0.79**.
- **Weekend Effect:** This strong correlation suggests that traffic behavior on weekends differs considerably from weekdays, likely due to changes in commuter patterns and recreational travel.

Incorporating these features will enhance the model's temporal sensitivity, allowing it to more accurately capture and predict these distinct patterns.

Weak Predictors: Hour and Month

Certain time-related features show minimal linear correlation with traffic volume:

- **Hour of Day:** The correlation coefficient for 'hour' is extremely low at **0.000072**.
- **Month of Year:** Similarly, 'month' shows a negligible correlation of **0.011**.

While their direct linear impact is minimal, these features might still play a role through more complex interactions or non-linear relationships, requiring deeper analysis or transformation to unlock their predictive power.

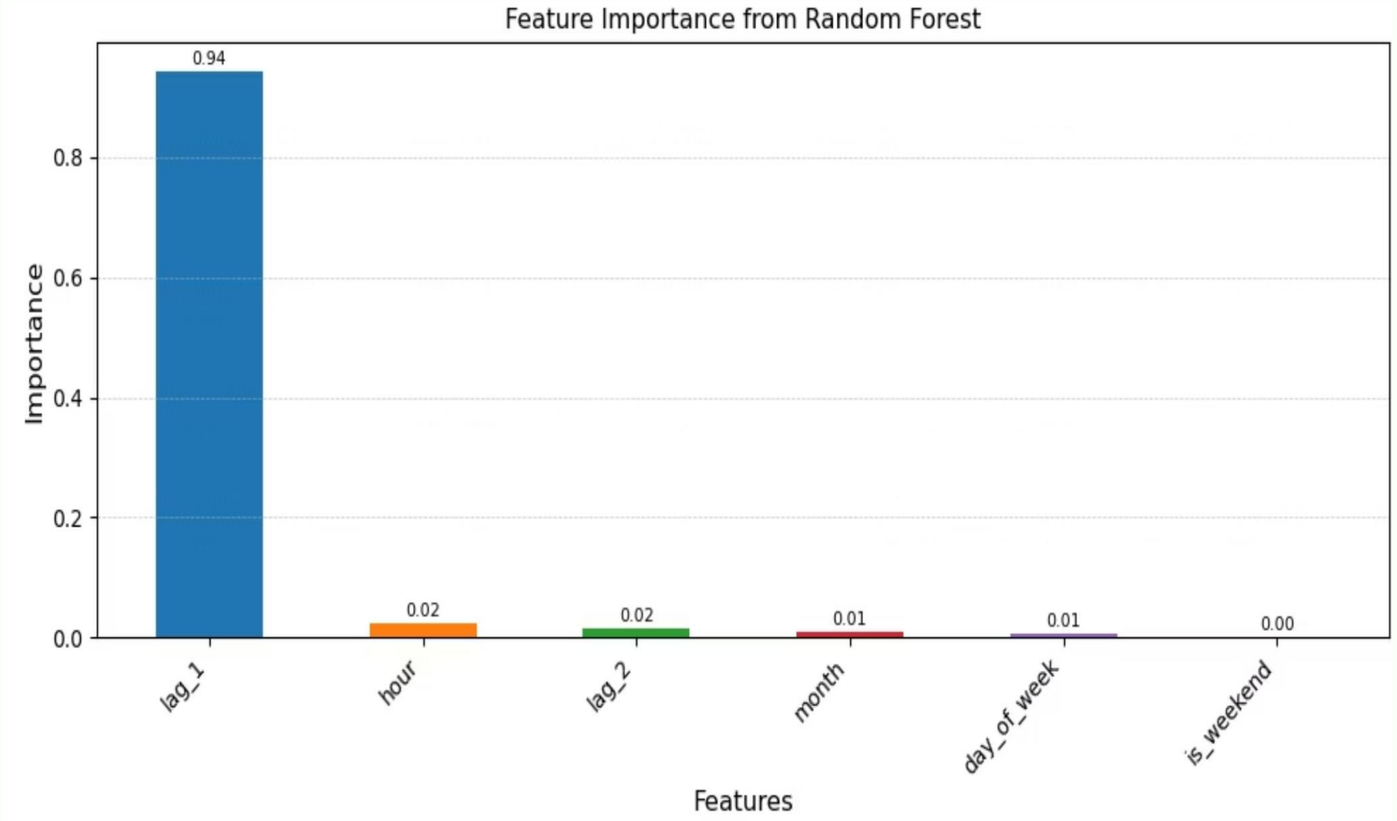
Negative Trends: Weekday Decline

Day of Week: Weak Negative Correlation (-0.084)

Our analysis revealed a weak negative correlation between the 'day_of_week' and traffic volume. This indicates a slight, albeit consistent, decrease in overall traffic volume as the week progresses towards the weekend.

While the correlation is minor, this trend suggests that daily routines and travel behaviors subtly shift throughout the week. Further exploration, possibly by segmenting data into specific time blocks (e.g., morning commute, midday, evening peak) or by individual junction types, could uncover more significant patterns and inform targeted traffic management strategies.

Random Forest Feature Importance



- The Random Forest model highlights the overwhelming dominance of 'lag_1' in predicting traffic volume, accounting for more than 80% of feature importance. Other features, including 'hour', 'month', 'day_of_week', and 'is_weekend', contribute minimally to the model's predictive power.
- This reinforces that future modeling efforts should primarily focus on 'lag_1' and potentially 'lag_2' for optimal performance, while other features might serve as secondary or interaction terms.

EDA-Driven Modeling Recommendations

1 Prioritize Lag Features

Focus model development on **lag_1** and **lag_2** given their strong predictive power, ensuring robust time-series handling.

2 Incorporate Weekend Indicators

Include **is_weekend** and **day_of_week** to capture distinct traffic patterns during non-business days.

3 Conditional Use of Hour/Month

Utilize **hour** and **month** only if their interaction with other features or non-linear patterns significantly improve model performance.

4 Robust Validation Strategy

Ensure model validation explicitly accounts for weekday and weekend splits to verify performance across varying traffic conditions.

Summary

Our exploratory data analysis reveals several key insights for building a robust traffic prediction model:

- **Lag Features Dominance:** Traffic volumes from previous hours (especially lag_1) are the most significant predictors.
- **Temporal Relevance:** Weekend indicators (is_weekend, day_of_week) are crucial for capturing different traffic behaviors.
- **Weak Predictors:** While hour and month have weak direct correlations, they may still offer value through complex interactions or transformations.
- **Random Forest Confirmation:** The Random Forest model independently validates lag_1 as the single most important feature for accurate predictions.